
Machine learning for Natural Language Processing

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Code disponible ici https://github.com/CaroLebrun/NLP_3A.git

Le travail sur les données a été réalisé avec Matéo Garbe.

Abstract

Our paper explores the correlation between candidates' professions and the topics emphasized in their campaign manifestos. Using a corpus of over 21 000 texts from the French legislative elections (1973–1993), we apply NLP methods to identify topics and analyse their distribution across socio-professional groups. Topics are extracted using Non-negative Matrix Factorization (NMF) on TF-IDF representations, while professions are grouped using semantic similarity and clustering techniques.

The results reveal systematic correlations between professions and political discourse: for example, candidates from education-related backgrounds are less likely to emphasize immigration, whereas those from private-sector professions focus more on fiscal and liberal economic issues. These results suggest that professional background plays a significant role in the ideas put forward in individual communication, even if these results could be improved by taking into account the party of the candidate that is a confounding factor.

1 Introduction and topic

Campaign manifestos ("professions de foi") constitute a key instrument of communication in legislative elections : they offer to all the candidates a given place to expose their policy priorities, ideological positions, and main propositions. Therefore, they offer a particularly adapted corpus for the comprehensive analysis of political discourse. Beyond their immediate electoral function, these texts are numerous enough to be used to discover patterns in them, including the influence of candidates' personal and professional backgrounds on their political positioning.

Investigating the relationship between professional background and the topics that are put forward is particularly relevant for several reasons :

- First, it contributes to the literature in political sociology and political communication by empirically examining whether and how social characteristics shape political discourse. While prior research has explored links between candidates' backgrounds and policy preferences, fewer studies have leveraged large-scale textual data to analyze these relationships in a systematic and quantitative manner
- Second, understanding such patterns help to explain the mechanisms of representation : if candidates from certain professions consistently insist on specific issues, it could indicate differentiated forms of issue advocacy and agenda-setting within the political field
- Finally, from a methodological point of view, this problem provides a relevant application case for Natural Language Processing (NLP), illustrating how computational techniques can be used to extract meaningful structure from unstructured political texts

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36 In this context, this project addresses the following research question: **to what extent is there a**
37 **relationship between a candidate's profession and the themes emphasized in their campaign**
38 **manifesto** ? More specifically, **do candidates with the same professional backgrounds systemati-**
39 **cally prioritize the same kind of topics?**

40 To answer this question, we analyse a corpus of campaign leaflets using NLP methods. We employ
41 topic modelling techniques to identify thematic structures within the texts and examine the distribution
42 of these themes across professional categories. NLP is also used to group professions into socio-
43 professional categories using two different approaches, that we will later explain.

44 Overall, this approach will enable us to explore potential correlations between socio-professional
45 characteristics and political topics and orientations ; it could provide a better understanding of how
46 individual backgrounds (through professions) may affect political topics in electoral context.

47 2 Presentation of the dataset

48 The Archelec corpus is a very rich collection of documents consisting of the manifestos of candidates
49 in the elections of the French fifth republic. Here we focus on legislative elections, to avoid any noise
50 from another election.

51 Therefore our total dataset contains 21 167 texts, divided between five legislative elections (1973,
52 1978, 1981, 1988, 1993). Many elements are available for each manifesto, such as name, age and
53 gender of the candidate, its political affiliation, its past political mandates, decoration, and so on.

54 The text of the manifesto often starts with "Sciences Po - Fonds Cevipof", which will require a small
55 first cleaning independently of the method later used to analyse the topics. We also use functions to
56 make sure our manifestos' texts are perfectly clean from any abnormal character that could disturb
57 later analysis.

58 Concerning the declared professions, there are three main issues with it :

- 59 • First, almost half of the manifestos have no profession ; therefore, we have only 11 702
60 manifestos that can be matched to a profession.
- 61 • Second, the number of stated professions is 3 397, but many of them could be almost identical
62 : since the candidates declared themselves their profession, there was no standardization,
63 and therefore we will have to find a way to group them.
- 64 • Third, we cannot analyse the profession at a individual levels : even with standardized
65 professions, there would still be too many of them to analyse this work at a larger level ; we
66 will need to create much smaller total number of professions to be able to conclude on our
67 results.

68 3 State-of-the-art

69 The idea of analyzing the link between a candidate's statement and the candidate's job is quite specific
70 and has not been much studied in the literature. Most previous works focus more on the relationship
71 between candidate background and their political ideas or ideological positions. In general, these
72 studies aim to understand how socio-demographic characteristics, such as profession or education,
73 can influence political opinions ([Pedersen et al., 2019] ; [Atkeson and Hamel, 2020] ; [Coffé and
74 Theiss-Morse, 2016]), but they do not directly study how this background is reflected in campaign
75 texts.

76 At the same time, the recent development and the increasing use of machine learning methods have
77 enlarged the set of tools available for political analysis. In particular, Natural Language Processing
78 (NLP) makes it possible to analyse large quantities of political texts in an automated way. We can
79 find different examples of the use of NLP to study political content, such as speeches, debates or
80 manifestos, using methods like topic modelling, sentiment analysis or similarity measures.

81 More specifically, NLP has already been applied to electoral manifestos. For example, Orellana and
82 Bisgin [2023] show that NLP can be used to analyse similarities between manifestos, to identify main

83 topics, and also to study sentiment. In short, NLP can be useful for a wide scope of political analysis.
84 In another context, [Bielik, 2020] analyse manifestos in Central and Eastern European countries and
85 show that social-democratic parties tend to have a stronger pro-European position, and that their
86 manifestos mainly focus on socio-economic issues.

87 However, we can see that most of these analyses are conducted at the party level : manifestos can be
88 seen as documents that represent the point of view of a party, and the objective is therefore to analyse
89 the topics or positions associated with this party ; but this approach does not take into account the
90 differences that may exist between individual candidates.

91 In this paper, we adopt a different approach by focusing on the individual level : we analyse
92 candidates' manifestos and we try to show a possible correlation between a candidate's profession
93 and the main topics of their manifesto. This allows us to go further than party-level analysis and to
94 better understand how individual characteristics may be linked to political discourse.

95 **4 Methodology**

96 The main structure of our analysis will be this one : first, creating clusters of texts based on topics
97 ; second, creating groups of professions to enable this profession-level analysis of the manifesto ;
98 finally, finding if there is a link between the manifesto and the profession of its candidate.

99 To get the topic of each candidate statement, we compared two methods :

- 100 • Non-negative Matrix Factorization (NMF) is particularly adapted to this study since it
101 produces interpretable and distinct topics based on term co-occurrence patterns. It relies on
102 TF-IDF, that will find discriminative terms across documents, while reducing the impact
103 of very frequent but less informative words ; it matters particularly for our manifestos, in
104 which certain generic expressions may appear across all texts. Therefore, thanks to TF-IDF
105 representations, NMF put forward discriminative words, which are essential to identify the
106 main topics in electoral manifestos. Moreover, NMF does not rely on strong probabilistic
107 assumptions and tends to generate more coherent and stable topics in practice, especially
108 for relatively structured political texts. This makes it more appropriate to comparing topics
109 across documents.
- 110 • BERTopic, which relies on BERT-based embeddings, captures semantic relationships beyond
111 simple word frequency. It means that the model can detect topics even when different words
112 are used to express similar ideas. Such contextual understanding is particularly relevant for
113 political discourse, since similar meanings can be expressed in different ways. However,
114 BERTopic depends on clustering in embedding space, which can lead to less stable or less
115 interpretable topics, especially in corpus with quite homogenous texts. Therefore, it is best
116 used as a complementary exploratory method.

117 Another option for topic modelling would have been to use LDA (Latent Dirichlet Allocation), that
118 relies on probabilistic assumptions. However, using these assumptions can limit its flexibility and
119 often lead to less discriminative topics compared to NMF. Moreover, LDA tends to not perform well
120 on datasets where clear lexical distinctions are important, as it is the case with candidates' manifestos.
121 In the end, it seemed less adapted to our subject.

122 The pre-processing of the texts depends on the representation used : for TF-IDF, the computations
123 are based on frequencies of word ; it therefore requires to lemmatize the words and remove all kinds
124 of punctuation. On the other hand, encoders such as Sentence-Camembert will use all available
125 information, such as the structure of the sentence ; this simplifies widely the treatment to apply on
126 the texts before clustering the texts to identify the topics.

127 About our second strategy, several encoders can be used, and we tried two of them :

- 128 • *paraphrase-multilingual-MiniLM-L12-v2*, which is light and adapted to all languages ;
129 however, this adaption can also impact the precision of the encoding, lacking some subtitles
130 of the language
- 131 • *sentence-camembert-base*, which is a specific French encoder, performing better for text in
132 French.

In theory, compared to multilingual or English-based models, the Camembert should capture linguistic nuances and context more effectively, leading to more accurate semantic embeddings and improving the quality of topic modelling with BERTopic. In our study, we did find that Camembert was slightly more effective (and more time-consuming), which justify its use here.

However, overall, TF-IDF combined with non-negative matrix factorization gave us better results, *i.e.* was able to dispatch the manifestos into more various categories, even if theoretically BERTopic is able to capture deeper semantic relationships ; here what matters is the interpretability of topics and their differentiations. We therefore use NMF for the rest of our analysis.

This choice could be questioned since we lack precise metrics to evaluate the "quality" of a topic, and we therefore rely on qualitative evaluation of the results from the two methods. We perform an indirect validation *infra* by comparing the topics with party affiliation, and check that there are consistent. It provides further evidence that our model captures meaningful structures for our topics.

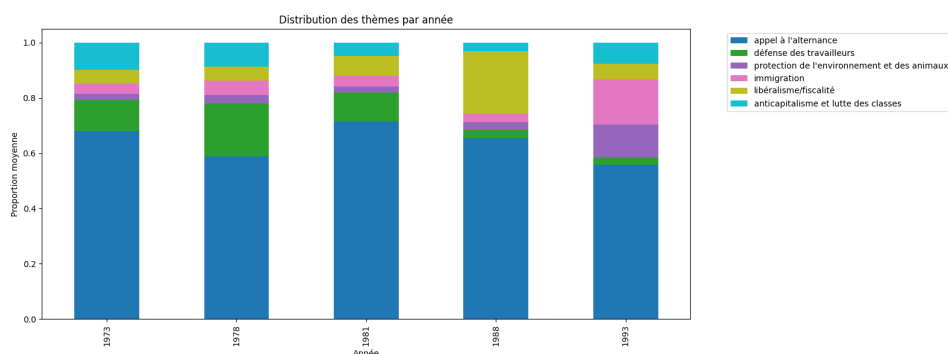


Figure 1: Distribution of topics per year

Another methodological challenge was to analyse the professions in the candidate's manifesto : the candidates just had to declare a profession, and sometimes they even have two professions (see above). It requires us to also perform a topic analysis on these jobs to group them into larger categories.

We tried two different methods to classify all the professions into smaller topics :

- First, we use embedding and clustering to assign candidates to a broader occupational category (named "PCS", which is the French nomenclature of socio-professional categories), at a very detailed level.

We use CamemBERT on professions and socio-professional categories, such that each of them is transformed into vectors in the same semantic space. Then we measure the cosine similarity between the vectors of the professions and those of the socio-professional categories. Using the similarity score, we can match each profession to its socio-professional category and then use this category as a group to perform the analysis linked to the topic of the manifesto.

- Second, we use again the same process as in the manifestos' topic clustering to cluster the professions in the same way. However, we use here k-means instead of DBScan, since DBScan only produces three categories even when adjusting the parameters, while k-means enables to fix ourselves the categories. We select empirically, through trying and testing, the number of categories, since there is a trade-off between getting precise and homogenous categories and interpretable categories. We finally decide to get 12 categories, which is the minimum number to get meaningful categories that are not putting together too different professions.

5 Results

5.1 The different topics

Using non-negative matrix factorisation, we get six main topics, that we have called :

- Appel à l’alternance
- Défense des travailleurs
- Protection de l’environnement et des animaux
- Immigration
- Libéralisme/fiscalité
- Anticapitalisme/lutte des classes

We then compare these topics to the political parties, to check if our results make sense.

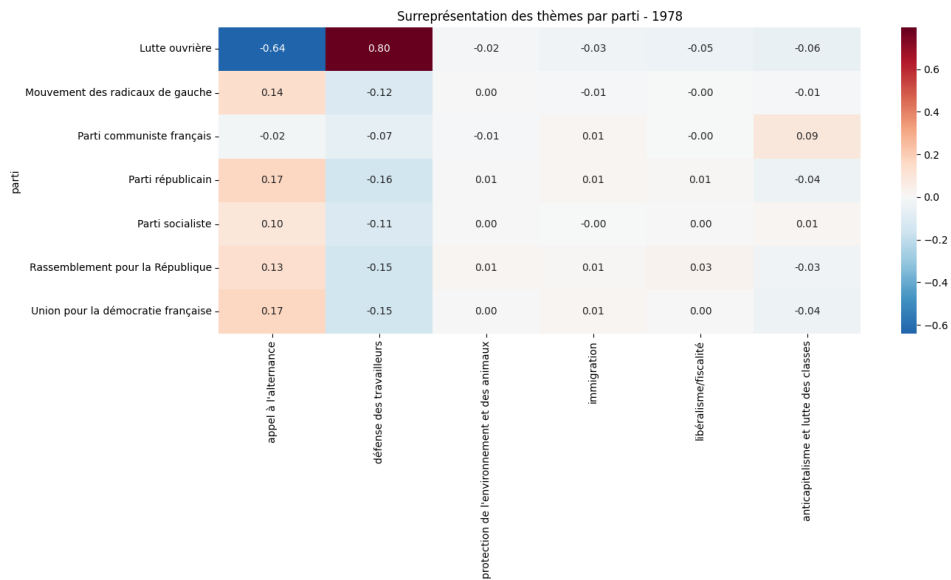


Figure 2: Over-representation of topics depending on political party of the candidate

Here for example, we see that "lutte ouvrière" is overrepresented in manifestos with "défense des travailleurs" being the main topics, which indeed makes sense.

5.2 The professions

Our two different analysis enables us to build two types of categories.

Our first method, with the socio-professional categories, gives us the following result (see figure 3)

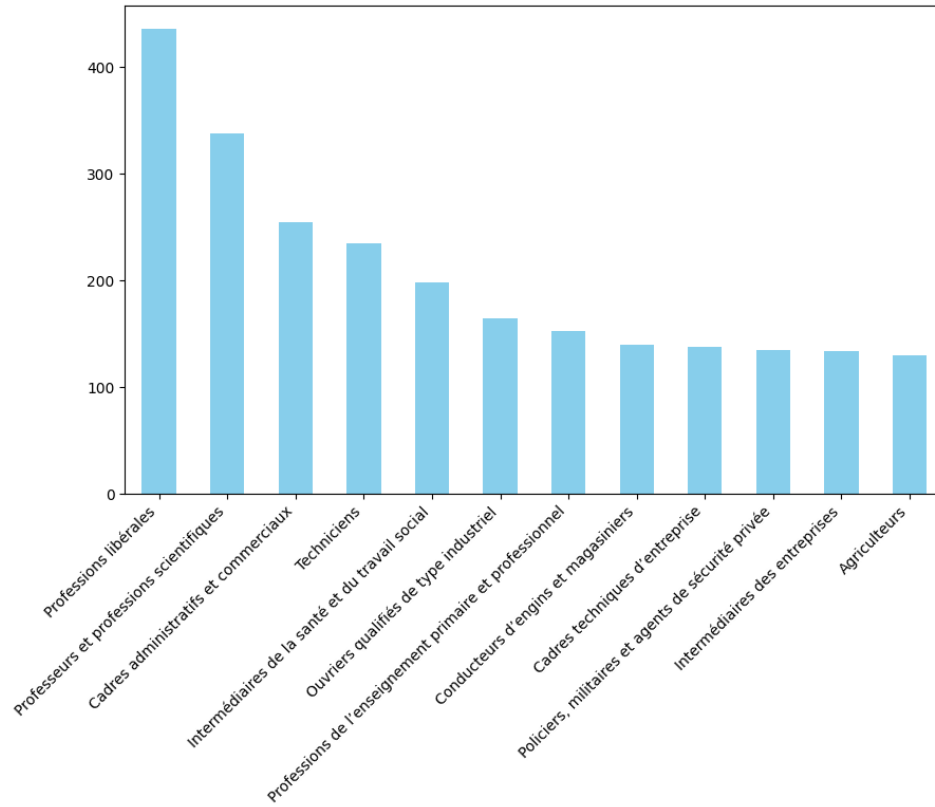


Figure 3: Distribution of the socio-professional categories

181 Our second method (see figure 4) gives slightly different results, since all the known professions are
 182 within the twelve clusters, while the previous graph was just about the most represented categories.

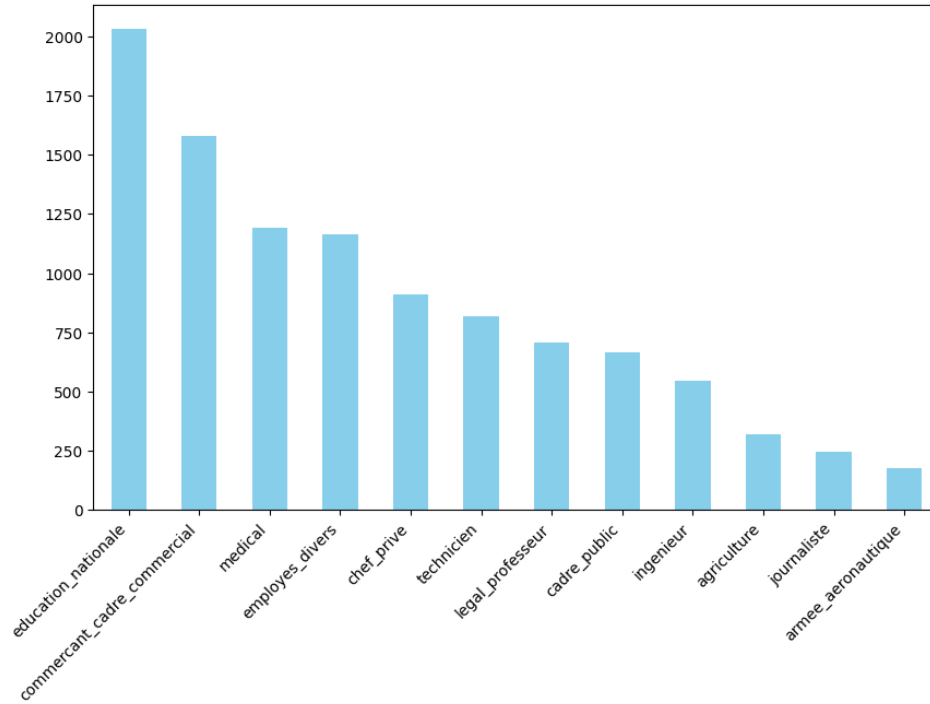


Figure 4: Distribution of the professions done in clusters

We see here that many candidates are teachers or workers in teaching field, followed by merchants or executives, then workers in medical field (such as doctors). This is coherent with the figure 3, since workers in medical field are often within "professions libérales", while teachers could be put in "professeurs et professions scientifiques", and merchants and executives belong to "cadres administratifs et commerciaux".

5.3 Correlation between professions and topics

We observe strong correlations between certain professions and the topics put forward in campaign manifestos. For instance, teachers (i.e., workers in the "éducation nationale") are much less likely to talk about themes related to immigration (see Figure 6). This result is consistent with empirical evidence, that suggest that higher levels of education are associated with a lower propensity to support far-right parties, which tend to focus more heavily on immigration issues Rynkiewicz et al. [2016].

Similarly, candidates from professions related to business ownership or the private sector are more likely to emphasize themes associated with "liberalism/fiscal policy" (see Figure 5). This may reflect their greater exposure to taxation and economic regulation, which could influence their political priorities.

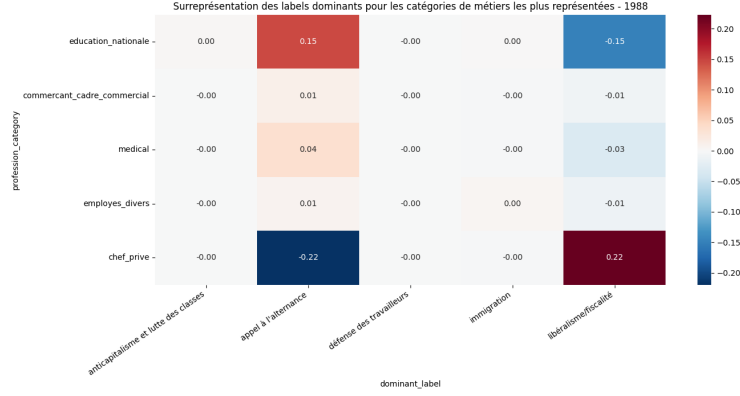


Figure 5: Over-representation of topics depending on the professions

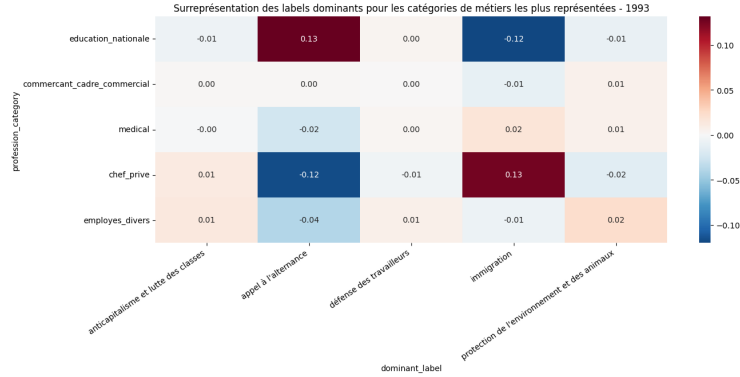


Figure 6: Over-representation of topics depending on the professions

We also perform a chi-square test of independence, that reveals a highly significant association between profession and dominant topic (value of chi-square is 1245.51, and the p-value is much smaller than 0.001). This result strongly rejects the null hypothesis (*i.e.* independence), indicating that the distribution of topics across professions is not random. To say it in another way, candidates' professional backgrounds are systematically associated with the topics they emphasize in their campaign manifestos.

This result supports the patterns observed in the heatmaps; it suggests that some professions are more likely to talk about specific topics in their manifestos. However, while the test confirms the existence of a relationship, it does not indicate its strength or direction.

6 Conclusion

This paper seeks to investigate whether candidates' professions are associated with the themes they put forward in their campaign manifestos. Using a large corpus of French legislative manifestos (1973–1993) and Natural Language Processing techniques, we identified major topics and analysed their distribution across socio-professional groups.

Our results provide strong evidence of a systematic relationship between profession and political discourse, with clear correlation seen in heatmaps and in the chi-square test of independence. It suggests that these patterns are unlikely to be driven by chance, and that certain professions tend to emphasize specific themes. Overall, this suggests that individual professional backgrounds play an important role in shaping political communication, independently of party-level dynamics.

However, several limitations should be acknowledged.

218 • First, a large share of manifestos does not include information on profession, which may
 219 introduce selection bias : if candidates from lower socio-professional background are more
 220 likely to have their profession missing, then the correlations we observed may not be
 221 generalized to the entire population of candidates.

222 • Second, the grouping of professions relies on several methodological choices that may
 223 influence the results (preferring NMF to BERTopic for example, or deciding the number
 224 of topics that is fixed *a priori*). Moreover, topics are interpreted based on their most
 225 representative words, and the labels we gave to the topics may not perfectly represent the
 226 clusters of topic (here for example, some themes overlap, and others could be only partially
 227 captured by the model). The method we choose ignore word order and syntactic structure,
 228 and this could have limited our ability to distinguish more complex patterns.

229 • Finally, while the analysis identifies associations, it does not establish causal relationships
 230 between professional background and political discourse : a major confounding factor could
 231 be political party affiliation. Parties indeed strongly structure political discourse, and since
 232 some professions could be overrepresented in some parties, the observed associations could
 233 partly reflect party effects rather than a direct effect of profession on manifestos.

234 This work could be therefore be extended, by using additional individual-level variables (for example
 235 education, gender, political experience), testing alternative methods for topic modelling and profession
 236 classification, or applying similar approaches to more recent electoral data (our dataset stops here at
 237 1993 legislative elections). More broadly, this study put forward the value of using textual analysis at
 238 a large level with socio-demographic information to better understand what shapes the topics used in
 239 political communication.

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